

WCT SERIES (10 TO 1,500 TR)

Industrial FRP Round & Square Cooling Towers

Aerodynamic Induced-Draft Counterflow & Crossflow Towers Engineered with Heavy-Duty Isophthalic FRP & High-Efficiency PVC Fills



10-1,500 TR

Round Bottle & Square
Modular Range

ISOPHTHALIC FRP

Corrosion & UV-Resistant
Reinforced Casing

ROTARY SPRINKLER

Self-Propelled Low-
Pressure Distribution

4°C APPROACH

Tight Approach to Design
Wet Bulb Temp

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WCT Series cooling towers provide evaporative heat rejection for industrial water chillers, plastic injection machines, diesel generator jackets, induction furnaces, and chemical plants. Available in two distinct architectural configurations—WCT-RL aerodynamic round bottle towers and WCT-SL modular square crossflow towers—each unit is constructed with premium isophthalic polyester resin reinforced with chopped strand fiberglass mats, ensuring complete resistance against corrosion, algae, and UV radiation.

Working Principle: Warm process water enters the tower and is distributed over high-efficiency virgin PVC honeycomb fills. In WCT-RL round towers, water is discharged through a calibrated multi-arm rotary alloy sprinkler head that rotates smoothly via jet reaction force at low water pressure (0.3 to 0.7 bar). In WCT-SL square towers, gravity distribution basins meter water through clog-free target nozzles. An induced-draft axial fan pulls dry ambient air upward through the falling water film, driving rapid evaporative cooling down to a tight 4°C approach to wet-bulb temperature.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Warm Condenser Water (37°C) → Inlet Riser → Self-Rotating Sprinkler / Gravity Basin → PVC Honeycomb Fill Media (Counterflow Air) → Cold Sump Basin (32°C) → Recirculation Pump → Chiller Condenser

Heavy-Duty Isophthalic FRP Casing

Precision molded with continuous chopped strand fiberglass and UV-inhibited isophthalic gelcoat. Impervious to acidic industrial atmospheres, coastal saline fog, chemical fumes, and biological growth with zero painting maintenance.

Self-Rotating Alloy Sprinkler System

Precision machined rotary sprinkler head rotates automatically on sealed stainless steel ball bearings via the dynamic reaction of water jets. Provides 360° uniform water distribution without requiring external motorized drives.

100% Virgin PVC Honeycomb Fills

Cross-fluted honeycomb PVC fills with integral drift eliminators and louver faces. Withstands continuous water temperatures up to 55°C (polypropylene option up to 80°C) with minimal airflow resistance and drift loss <0.005%.

Direct-Driven Low-RPM Aero-Foil Fans

Balanced cast aluminum or FRP aerofoil blade fans driven directly by weather-proof IP55 flange-mounted electric motors, eliminating maintenance-heavy belts, pulleys, and gearboxes.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Nominal Flow: 0.6 m³/hr per TR (3.0 GPM/TR) | Hot Water In: 37°C | Cold Water Out: 32°C | Wet Bulb Temp: 28°C (Approach 4°C) | Casing: UV-Resistant FRP | Fill: PVC Cross-Flute

MODEL CODE	NOMINAL TR	WATER FLOW M ³ /HR	MOTOR HP	FAN RPM	ROUND DIA × H (MM)	SQUARE L×W×H (MM)	OPER WT KG
WCT 010	10 TR	6 m ³ /hr	0.5 HP	1440 RPM	Ø 930 × 1690	930 × 930 × 1690	280 kg
WCT 015	15 TR	9 m ³ /hr	0.5 HP	1440 RPM	Ø 1050 × 1700	1050 × 1050 × 1700	340 kg
WCT 020	20 TR	12 m ³ /hr	0.75 HP	1440 RPM	Ø 1150 × 1750	1150 × 1150 × 1750	390 kg
WCT 025	25 TR	15 m ³ /hr	0.75 HP	1440 RPM	Ø 1250 × 1750	1250 × 1250 × 1750	450 kg
WCT 030	30 TR	18 m ³ /hr	1.0 HP	960 RPM	Ø 1350 × 1850	1350 × 1350 × 1850	550 kg
WCT 040	40 TR	24 m ³ /hr	1.0 HP	960 RPM	Ø 1500 × 1950	1500 × 1500 × 1950	680 kg
WCT 050	50 TR	30 m ³ /hr	1.5 HP	960 RPM	Ø 1650 × 2150	1650 × 1650 × 2150	850 kg
WCT 060	60 TR	36 m ³ /hr	2.0 HP	960 RPM	Ø 1800 × 2250	1800 × 1800 × 2250	1,050 kg
WCT 075	75 TR	45 m ³ /hr	2.0 HP	960 RPM	Ø 2000 × 2350	2000 × 2000 × 2350	1,250 kg
WCT 100	100 TR	60 m ³ /hr	3.0 HP	960 RPM	Ø 2250 × 2500	2250 × 2250 × 2500	1,650 kg
WCT 125	125 TR	75 m ³ /hr	5.0 HP	960 RPM	Ø 2500 × 2650	2500 × 2500 × 2650	2,100 kg
WCT 150	150 TR	90 m ³ /hr	5.0 HP	960 RPM	Ø 2750 × 2800	2750 × 2750 × 2800	2,450 kg
WCT 200	200 TR	120 m ³ /hr	7.5 HP	720 RPM	Ø 3150 × 3100	3150 × 3150 × 3100	3,200 kg
WCT 300	300 TR	180 m ³ /hr	10.0 HP	720 RPM	Ø 3800 × 3500	3800 × 3800 × 3500	4,800 kg
WCT 500	500 TR	300 m ³ /hr	15.0 HP	720 RPM	Ø 4800 × 4100	4800 × 4800 × 4100	7,800 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- Square modular towers (WCT-SL) can be banked in multi-cell configurations for capacities up to 1,500 TR.
- High-temperature PP fills available for process water applications entering up to 80°C (e.g. steel furnaces, quenching tanks).
- All structural hardware fabricated from Hot-Dip Galvanized (HDG) steel or SUS304 stainless steel.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

- Plastic Processing & Blow Molding** Cools hydraulic oil exchangers, barrel throat jackets, and chiller condensers for reliable 24/7 continuous production.
- Steel Rolling & Induction Melting** Protects water-cooled power cables, crucible induction coils, and SCR thyristor panels from destructive overheating.
- HVAC Central Water-Cooled Chillers** Rejects building heat from centrifugal and screw chiller condensers at lowest electrical kW/TR consumption.
- Diesel Generator & Air Compressor Jacketing** Maintains engine and compressor aftercoolers at optimal thermal operating conditions, extending cylinder liner life.
- Chemical & Distillation Condensers** Dissipates high condensing heat loads across distillation columns, solvent recovery stills, and batch autoclaves.

Engineering Comparison: Win WCT vs. Standard Commercial Cooling Towers		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
FRP Structure	High-density isophthalic resin with embedded UV stabilizers	General-purpose orthophthalic resin prone to chalking & pinholes
Water Sprinkler	Machined aluminum/brass rotary sprinkler on sealed bearings	Plastic sprinkler head that seizes within months
Drift Loss	Integral cellular drift eliminators restrict loss <0.005%	Open spray pattern causing >0.05% water carryover
Structural Framing	Heavy HDG steel structural chassis with SS304 fasteners	Lightweight painted angle iron that rusts in industrial air

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- Dynamic balancing of fan assembly to ISO 1940 Grade G2.5 for vibration-free running
- Sprinkler head flow calibration and rotation velocity test at minimum 0.3 bar pressure
- Hydrostatic leak test on FRP cold water basin prior to factory dispatch
- Full load motor running current and insulation Megger verification

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 12 Months comprehensive warranty on structure, motor, and internal components. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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